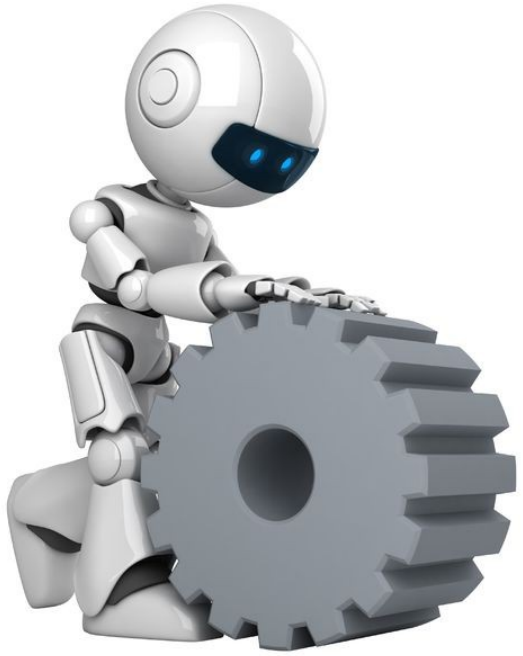
ICAT3110

Intelligent Robotics



Lecture 3:
Wheeled Locomotion

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University of Vaasa

Reminder again …

Schedule:

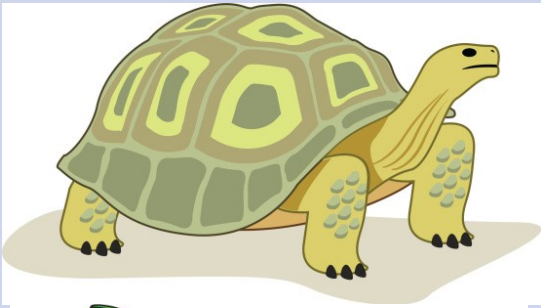
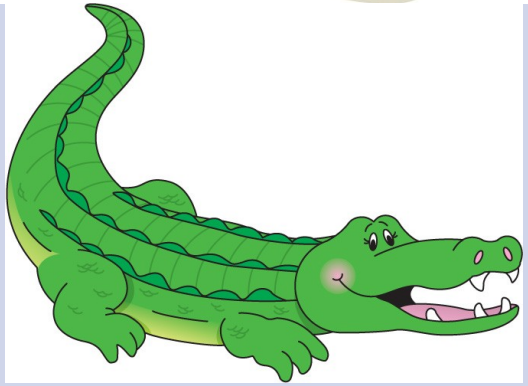
Note that we always start **+15 minutes** past the lecture time to allow more participants to join!
E.g., lecture time at 14:00, the actual start at 14:15

Lecture No.	Date	Time	Location	Zoom Link
Lecture 1	26/1/2026	14:00-16:00	C303 Tervahovi	https://uwasa.zoom.us/j/63930259473?pwd=bhZzKyIc5hTG5TMDtEny6AEtCG1Hu2.1
Lecture 2	29/1/2026	14:00-16:00	C303 Tervahovi	https://uwasa.zoom.us/j/68780511364?pwd=dCbvCocQj2bWNhki8NS7eTSnFcVJza.1
Lecture 3	02/2/2026	10:00-12:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/64553376642?pwd=DTsPD5AhDL4DRZHu8WVEsw4avb2aQi.1
Lecture 4	04/2/2026	14:00-16:00	C203 Kurten Tervahovi	https://uwasa.zoom.us/j/65745013217?pwd=oosU6dvtnKDaasrHhu3o4aRjpypbIc.1
Lecture 5	10/2/2026	12:00-14:00	C203 Kurten Tervahovi	https://uwasa.zoom.us/j/67071083834?pwd=PtK77a3QSNVGPYwDYbx4IlYuzktMmB.1
Lecture 6	12/2/2026	10:00-12:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/64870927206?pwd=joc4JMVuy713ha7CKrkuCySOHW6qz3.1
Lecture 7	16/2/2026	10:00-12:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/62551103459?pwd=9E4O21JaoUuV8KqNIya2PbziYvzdw2.1
Lecture 8	18/2/2026	Group (A) 09:00-11:00 Group (B) 12:00-14:00	ABB Factory, Vaasa	**ABB Oy Factory Site Visit (No online meeting!)**
Lecture 9	23/2/2026	12:00-14:00	F118 Fabriikki	https://uwasa.zoom.us/j/66579698602?pwd=sXvOvxfa5mXyclqRMpW6B9w3u6N8uU.1
Lecture 10	27/2/2026	10:00-12:00	C303 Tervahovi	https://uwasa.zoom.us/j/62280835656?pwd=uhsz5NDoAr5GloKt0njvRaVVwfLRf9.1
Lecture 11	02/3/2026	12:00-14:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/61669450219?pwd=u75DuOhiuFsyJgVLFjRpmtaDyRnxTV.1
Lecture 12	05/3/2026	14:00-16:00	F118 Fabriikki	https://uwasa.zoom.us/j/66549239286?pwd=3Mhe9xqa9TRU0UuKGtRRmRBE9zk7DT.1

Recall !
Legged Locomotion

Leg Configuration and Stability

- Legged Robots are biologically inspired, of course these systems are ***successful***.

Two or Four Legged Mammals	Four Legged Reptiles	Six Legged Insects
	 	

Leg Configuration and Stability

- For a Mobile Robot with ***k*** legs,

The total number of possible events ***N*** for a walking machine is given by the relation

$$N = (2k - 1)\,!$$

- Hence for a biped walker *k = 2,* there are 6 possible events

1	2	3	4	5	6
Lifting Right leg	Lifting Left leg	Release Right leg	Release Left leg	Lift both Legs	Release both Legs

- **Question:** For a Mobile Robot with 6 Legs

$$N = (12 - 1)! = 11!$$

39 916 800 possible event !

Legged Mobile Robots

- It characterizes by series of contact-points between the robot and ground.
- **Advantages**: Adaptability and Maneuverability in rough environment.
- Because quality of ground does not matter in such case as long as Robot maintain contact with ground through number of its points.
- **One more Advantage:** Can manipulate objects with great skill.
- **Disadvantages:**
  - Power and Mechanical complexity.
  - The leg, which include degrees of freedom, must be capable of sustaining part of the robot's total weight, and in many robots must be capable of lifting and lowering the robot.

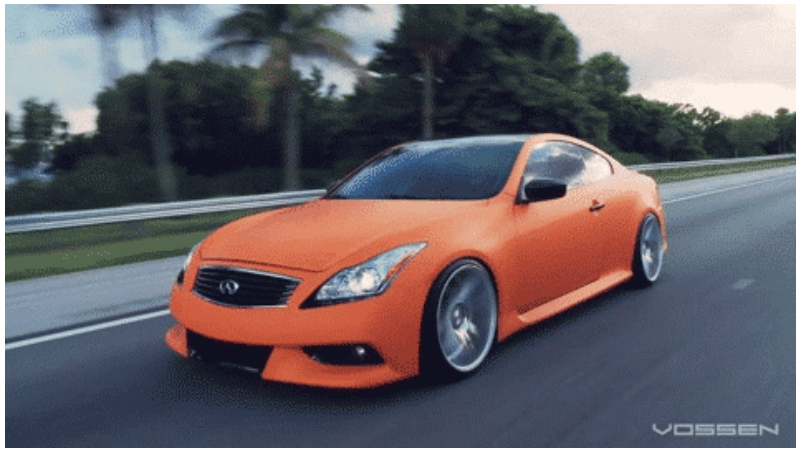
Those were Legged Robots

How about

Wheeled Mobile Robots?
?

Wheeled Mobile Robots

- By far it is the ***most popular locomotion*** mechanism in mobile robotics/vehicles designed by Human.



- Simple mechanism, very efficient in solid flat surface and keeps stability.
- Wheels are in **direct contact with ground** all the time so balancing is kept constant.
- **Three wheels** are sufficient to guarantee stable balance.
- **Two wheels** can also be stable BUT more effort and more control are needed.

Wheeled Mobile Robots

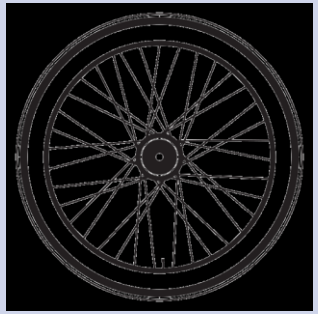
- Instead of worrying about balance, wheeled robot research tends to focus on the problems of **traction** and **stability**, **maneuverability**, and **control.**

- Can the Robot Wheels provide **sufficient traction and stability** for the robot to cover all of the **desired terrain?**
  Does the robot's wheeled configuration enable sufficient control **over the velocity of the robot**?
- There is a very large space of **possible wheel configurations** when considering possible techniques for mobile robot locomotion.

- **Firstly**, studying different **wheel types** with specific strengths and weaknesses.
- **Secondly**, examine complete **wheel configurations** that deliver particular forms of locomotion for a mobile robot.

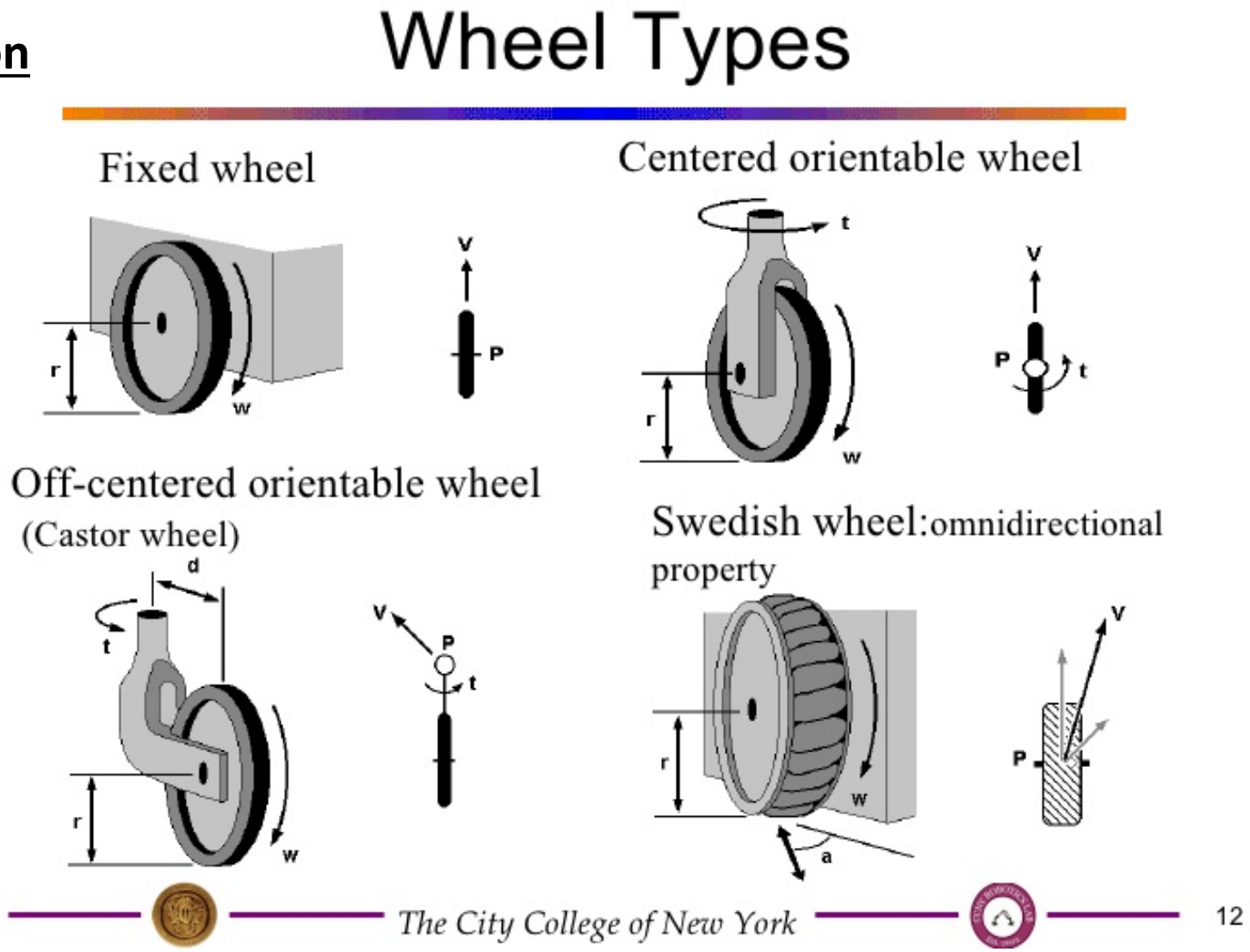
Wheeled Design

- There are **Four** major wheel classes.
- They differ widely in their kinematics, and therefore the **choice of wheel type** has a large effect on the **overall kinematics** of the mobile robot.

Standard Wheel	Castor Wheel	Swedish Wheel	Spherical Wheel
			
2 DOF Rotation around axis And contact point	2 DOF Rotation by steering joint And contact point	3 DOF **Video ….** Rotation around axis Rollers And contact point	**DOF:** Omni Directional **Ex:** Computer Mouse

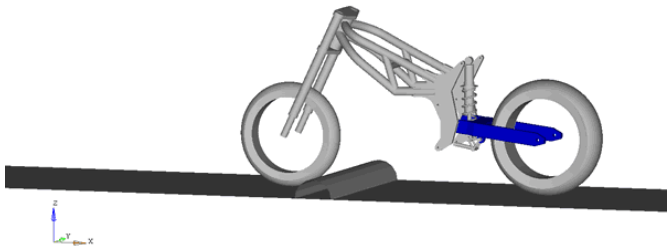
Wheeled Design

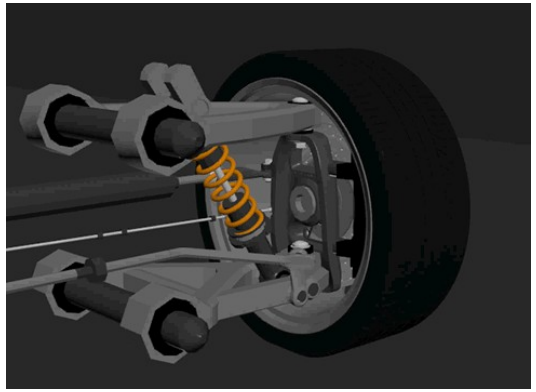
- **Another Classification**

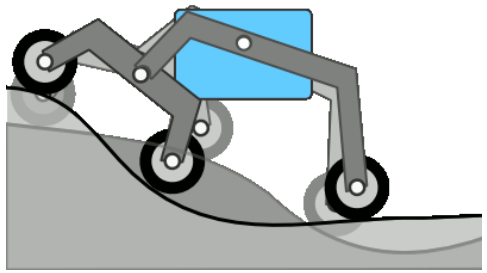


Wheeled Design

- For Robots which are designed for **all-terrain environments** and for robots with more than three wheels, a suspension system is normally required to **maintain wheel contact** with the ground.
- One of the simplest approaches to suspension is to design flexibility into the wheel itself. **Ex:** Manufacturers have applied a **deformable tire of soft rubber** to the wheel to create a primitive suspension.







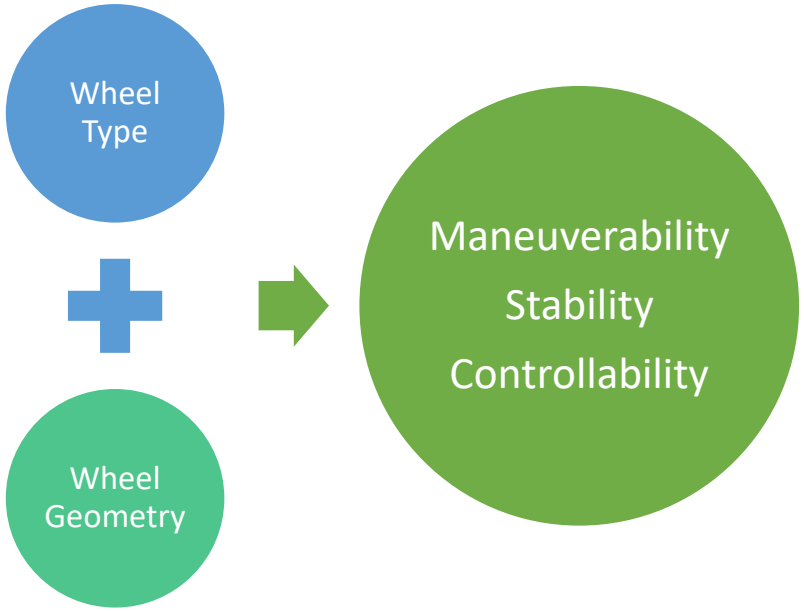
- Of course there should be more **sophisticated suspension** system in applications where the robot needs a more **dynamic suspension** for significantly **non-flat** terrain.

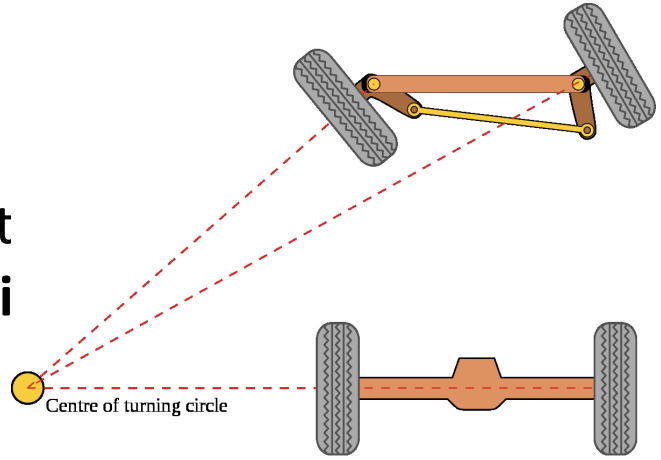
Wheeled Geometry

- Automobiles are designed for a highly standardized environment (**the road network**), mobile robots are designed for applications in a wide **variety** of situations.

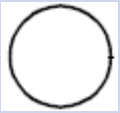
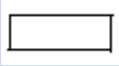
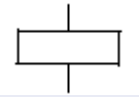
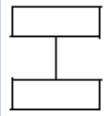
- Automobiles all share similar wheel configurations because there is **a region(s)** in the design space that maximizes <u>maneuverability</u>, <u>controllability</u> and <u>stability</u> for their standard environment: **the paved roadway.**

- The **Ackerman wheel** configuration of automobiles is not suitable for mobile Robots due to its **poor maneuverabili** unless the Robot is designed for road system as like cars.

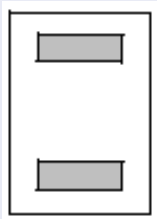
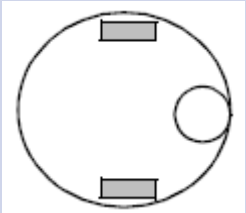
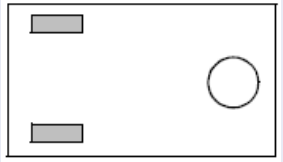




Wheeled Geometry

Icon	Description
	Unpowered omnidirectional wheel (spherical, castor, Swedish)
	Motorized Swedish wheel (Stanford wheel)
	Unpowered standard wheel
	Motorized standard wheel
	Motorized and steered castor wheel
	Steered standard wheel
	Standard connected wheels.

Wheeled Geometry

Number of Wheels	Arrangement	Description	Examples
2		One front **steering** wheel One rear **traction** wheel	Bicycle Motorcycle
		Two-wheel differential drive	Personal Cye
3		Two-wheel centered differential drive with a third point of contact	smartRob, EPFL
		Two independently driven wheels in the rear/front, 1 unpowered omnidirectional wheel in the front/rear	Pygmalion Many indoor robots

More ….

Wheeled Locomotion Stability

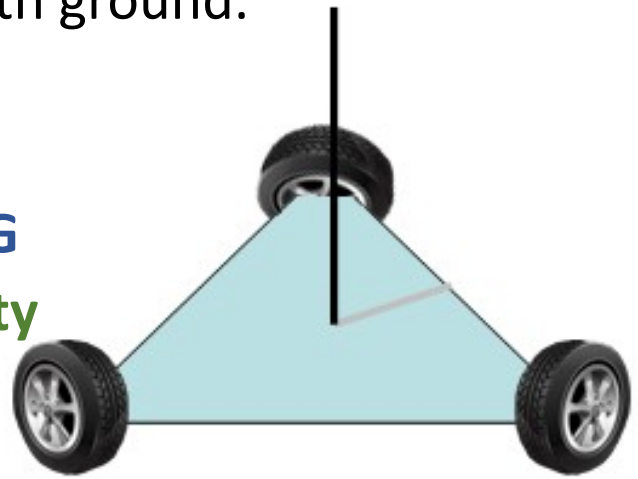
- Surprisingly, the minimum number of wheels required for **static stability is two**.
  <u>If the application is relatively small-sized and simple.</u>
- But in larger diameter wheels applications + High motor torque force.
  <u>Minimum Three wheels for static stability.</u> **And ?**
- Besides the number of wheels used, another important parameters is needed for static stability... **<u>Center of Gravity</u>**
- Center of Gravity should act within the contained triangle that is formed from contact points with ground.

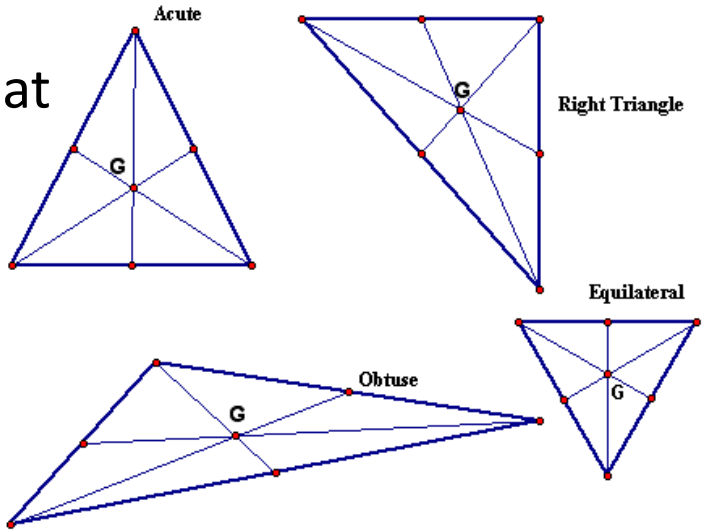
Irregular shapes

Non-uniform weights

Causes shifts in COG

Adding more wheels boosts Stability

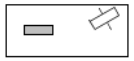
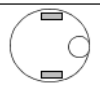
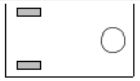
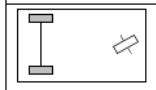
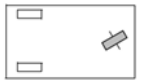
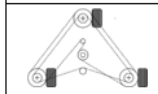




Let’s continue

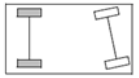
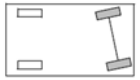
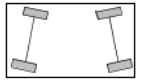
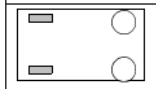
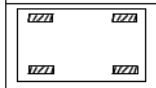
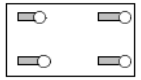
Wheeled Robots

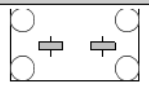
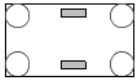
Wheeled Geometry (Check attached document)

# of wheels	Arrangement	Description	Typical examples
2		One steering wheel in the front, one traction wheel in the rear	Bicycle, motorcycle
		Two-wheel differential drive with the center of mass (COM) below the axle	Cye personal robot
3		Two-wheel centered differential drive with a third point of contact	Nomad Scout, smartRob EPFL
		Two independently driven wheels in the rear/front, 1 unpowered omnidirectional wheel in the front/rear	Many indoor robots, including the EPFL robots Pygmalion and Alice
		Two connected traction wheels (differential) in rear, 1 steered free wheel in front	Piaggio minitrucks
		Two free wheels in rear, 1 steered traction wheel in front	Neptune (Carnegie Mellon University), Hero-1
		Three motorized Swedish or spherical wheels arranged in a triangle; omnidirectional movement is possible	Stanford wheel Tribolo EPFL, Palm Pilot Robot Kit (CMU)
		Three synchronously motorized and steered wheels; the orientation is not controllable	"Synchro drive" Denning MRV-2, Georgia Institute of Technology, I-Robot B24, Nomad 200

Locomotion **35**

Table 2.1

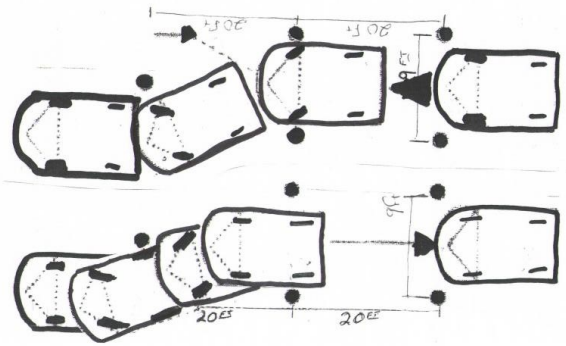
# of wheels	Arrangement	Description	Typical examples
4		Two motorized wheels in the rear, 2 steered wheels in the front; steering has to be different for the 2 wheels to avoid slipping/skidding.	Car with rear-wheel drive
		Two motorized and steered wheels in the front, 2 free wheels in the rear; steering has to be different for the 2 wheels to avoid slipping/skidding.	Car with front-wheel drive
		Four steered and motorized wheels	Four-wheel drive, four-wheel steering Hyperion (CMU)
		Two traction wheels (differential) in rear/front, 2 omnidirectional wheels in the front/rear	Charlie (DMT-EPFL)
		Four omnidirectional wheels	Carnegie Mellon Uranus
		Two-wheel differential drive with 2 additional points of contact	EPFL Khepera, Hyperbot Chip
		Four motorized and steered castor wheels	Nomad XR4000

# of wheels	Arrangement	Description	Typical examples
6		Two motorized and steered wheels aligned in center, 1 omnidirectional wheel at each corner	First
		Two traction wheels (differential) in center, 1 omnidirectional wheel at each corner	Terregator (Carnegie Mellon University)

Icons for the each wheel type are as follows:	
	unpowered omnidirectional wheel (spherical, castor, Swedish);
	motorized Swedish wheel (Stanford wheel);
	unpowered standard wheel;
	motorized standard wheel;
	motorized and steered castor wheel;
	steered standard wheel;
	connected wheels.

Wheeled Locomotion Maneuverability

- Maintaining movement in a strategic situation in a competitive environment to avoids a worse situation.
- Some robots are **omnidirectional**, meaning that they can move at any time in any direction along the ground plane (x,y) **regardless of the orientation** of the robot.



- This level of maneuverability requires wheels that can move in more than just one direction, and so **omnidirectional robots** usually employ **Swedish or spherical** wheels that are powered.

- **<u>As Example</u>**: Uranus-Carnegie Mellon.
- An **omni-directional** mobile robot.
- Using **Mecanum** wheels from Sweden

Wheeled Locomotion Maneuverability

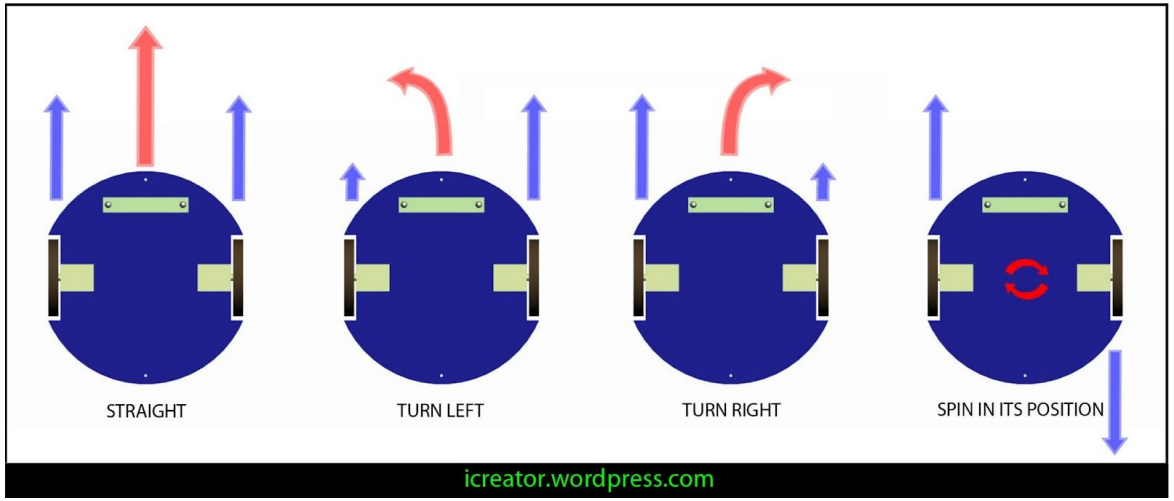
- In general, **the ground clearance** of robots with **Swedish** is **somewhat limited** due to the mechanical constraints of constructing omnidirectional wheels.
- As a solution, The **four-castor wheel** configuration can be installed in which each castor wheel is actively steered and actively translated.
- In this configuration, the robot is truly omnidirectional because, even if the castor wheels are facing a **direction perpendicular** to **the desired direction of travel**, the robot can still move in the desired direction by steering these wheels.
- Because the **vertical axis is offset** from the ground-contact path, the result of this steering motion is **robot motion**.

Wheeled Locomotion Maneuverability

- Higher maneuverability can be achieved only slightly inferior to that of the omnidirectional configurations.
- In such robots, motion in a particular direction may initially require **a rotational motion**.
- With a circular chassis and an axis of rotation at the center of the robot, such a robot **can spin** without changing its **ground footprint**.

- The most popular such robot is the **two-wheel differential-drive** robot where the two wheels **rotate** around the center point of the robot.

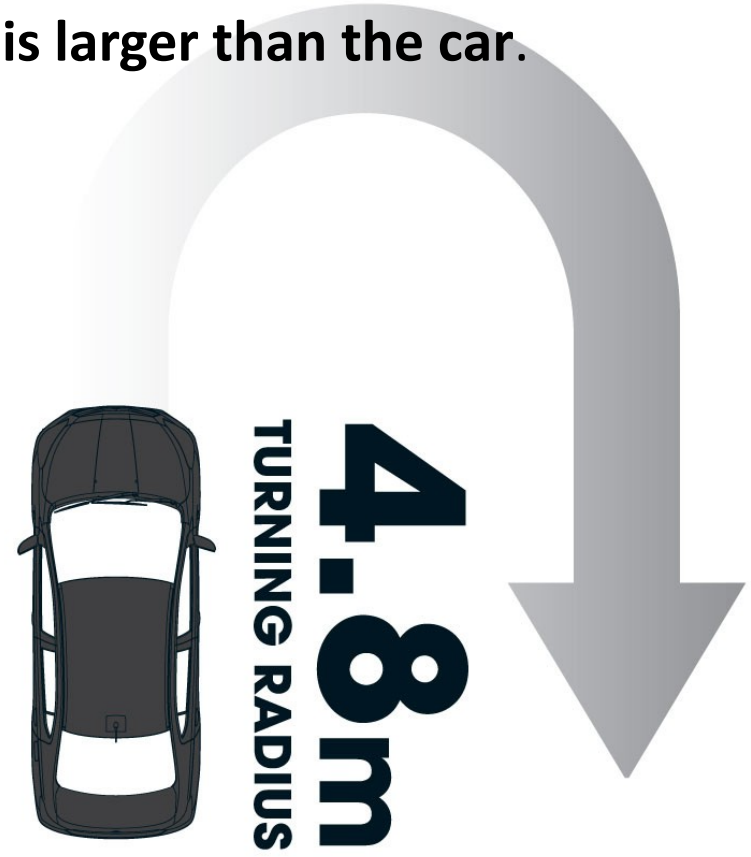
- One or two additional **ground contact** points may be used for **more stability**, based on the **application** specifics.



Video …

Wheeled Locomotion Maneuverability

- In contrast to the previous configurations, consider the **Ackerman steering configuration** which is common in automobiles.
- Such a vehicle typically has a **turning diameter that is larger than the car**.
- Furthermore, for such a vehicle to move **sideways** requires a **parking maneuver** consisting of **repeated** changes in direction **forward** and **backward**.
- Though, Ackerman steering geometries have been especially **popular** in the **hobby robotics** market **e.g.** Race Car kits.
- **One important advantage**: its directionality and steering geometry provide it with very **good lateral stability** in **high-speed** turns.



Wheeled Locomotion Controllability

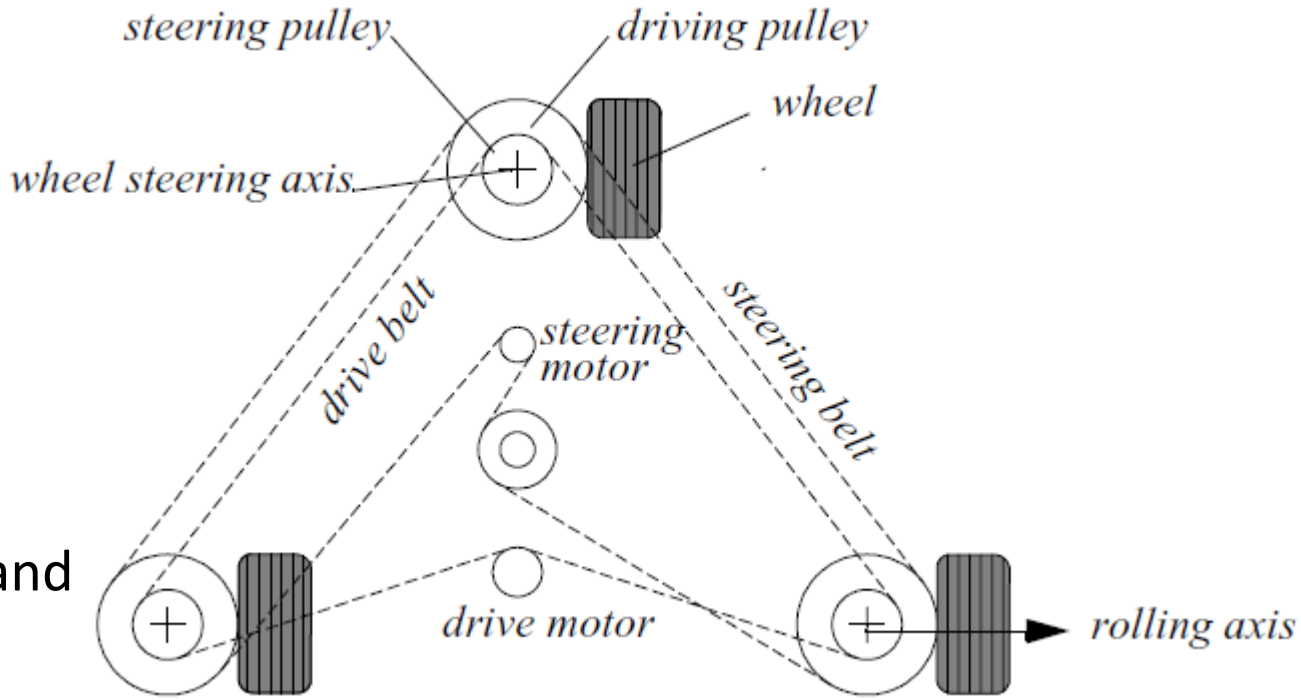
- There is generally an **inverse correlation** between **controllability** and **maneuverability**.
- For example, the omnidirectional designs (such as the four-castor wheel configuration) require significant processing **to convert desired rotational** and translational velocities to **individual wheel commands**.

- Controlling an omnidirectional robot for a specific direction of travel is also **more difficult** and often **less accurate** when compared to less maneuverable designs.
- For example, an Ackerman steering vehicle can **go straight** simply by **locking** the **steerable** wheels and driving the drive wheels.

- In summary, there is **no "ideal" drive configuration** that simultaneously maximizes stability, maneuverability, and controllability.
- Each mobile robot application **places unique constraints** on the robot **design problem**, and the **designer's task** is to choose the most appropriate drive configuration possible from among this space of compromises.

Wheeled Locomotion Case Study Synchro drive

- The synchro drive configuration is a **popular** arrangement of wheels in **indoor mobile robot** applications.
- It is an interesting configuration because, although there are **three driven** and steered wheels, only **two motors** are used in total.
- The one **translation motor** sets the **speed** of **all** three wheels together, and the **one steering** motor **spins** all the wheels together about each of their individual vertical steering axes.



Video ….

- Commercial research robots such as the **Nomadics 150** or the **RWI B21r** have been sold with this configuration.

Wheeled Locomotion Case Study

3 Spherical wheels

- Each actuated by **one motor**, the spherical wheels are suspended by three contact points, two given by spherical bearings and one by a wheel connected to the motor axle.
- This concept provides **excellent maneuverability** and is simple in design.
- Limited **to flat surfaces** and **small loads**.

Omnidirectional drive

4 Swedish wheels

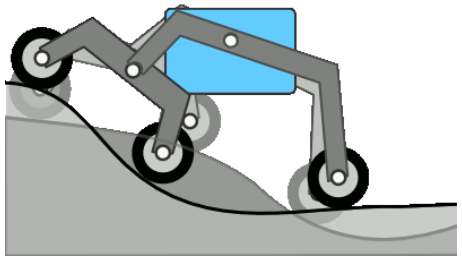
- Has been used **successfully** on several research robots, including the Carnegie Mellon **Uranus**.
- 4 Swedish 45-degree wheels, each driven by a separate motor.
- By varying the **direction of rotation** and **relative speeds** of the 4 wheels, the robot can be moved/spinned along any trajectory.

Wheeled Locomotion Case Study

Tracked slip/skid

- An alternative form of steering, termed **slip/skid**, may be used to reorient the robot by spinning wheels
- Robots that make use of tread have much larger ground contact patches that can **improve maneuverability** however changing the orientation of the robot usually requires a skidding turn.
- **The disadvantage** of the large amount of skidding during a turn, the exact center of rotation of the robot is hard to predict.

Wal



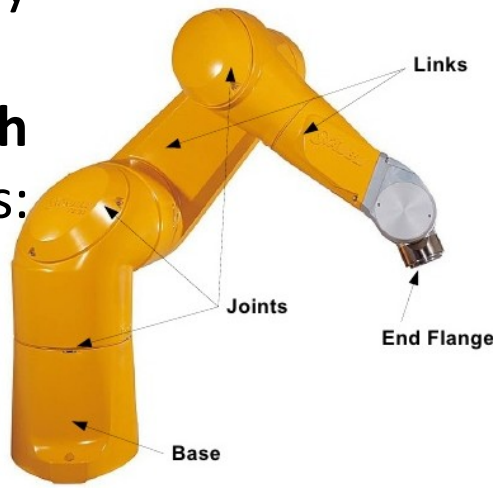
- Offers the **best maneuverability** in rough terrain. However, they are **inefficient on flat ground**.
- Hybrid solution, combining the **adaptability of legs with the efficiency of wheels**, offer an interesting compromise.
- Some Walking wheeled robots are capable of **climbing objects** up to two times its wheel diameter.

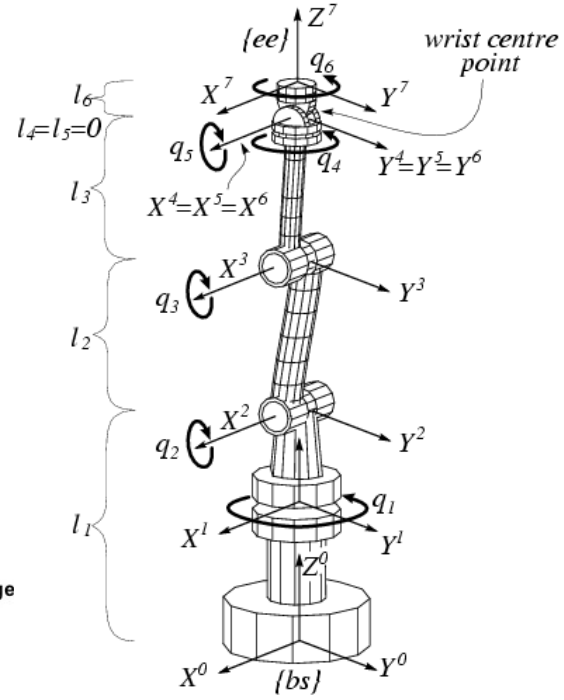
Locomotion (concluded)

Let’s talk about “Kinematics”

Kinematics

- Kinematics is the most **basic study** of how mechanical systems **behave**.
- In mobile robotics, we need to **understand** the **mechanical behavior** of the robot both in order to **design appropriate** mobile robots for tasks and to understand how to create **control software** for an instance of mobile **robot hardware**.
- Mobile robots are **not the first complex** mechanical systems to require such analysis. **Robot manipulators** have been the subject to Kinematics intensive study for more than thirty years.
- Sometime, manipulator robots are **much more complex** than **early mobile** robots: a standard welding robot may have **five or more joints**, whereas **early mobile** robots were simple differential-drive machines.

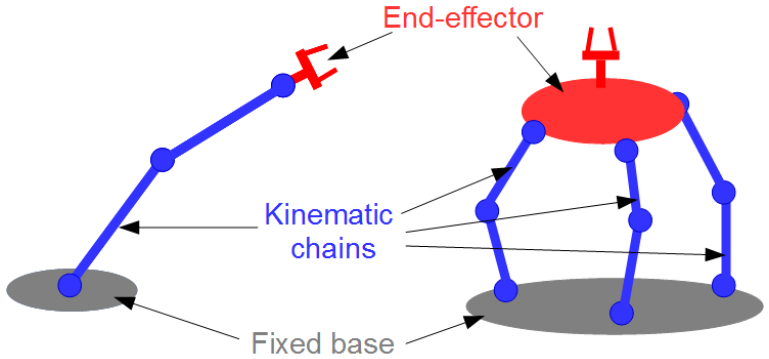


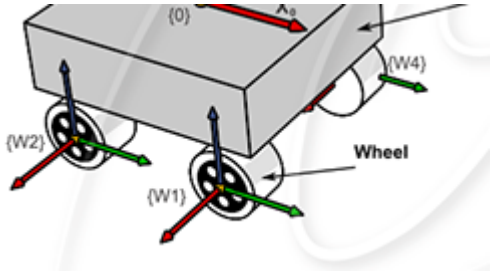


Kinematics

- A **manipulator** robot's **workspace is crucial** because it defines the **range of possible positions** that can be achieved by its **end relative** to **its fixture** to the environment.
- A **mobile** robot's **workspace** is **equally important** because it defines the **range of possible poses** that the mobile robot can **achieve** in its environment.

- The robot **arm's controllability** defines the **manner** in which **active** engagement of **motors** can be used to move from pose to pose in the **workspace**.
- Similarly, a mobile robot's **controllability** defines **possible paths** and trajectories in its workspace.



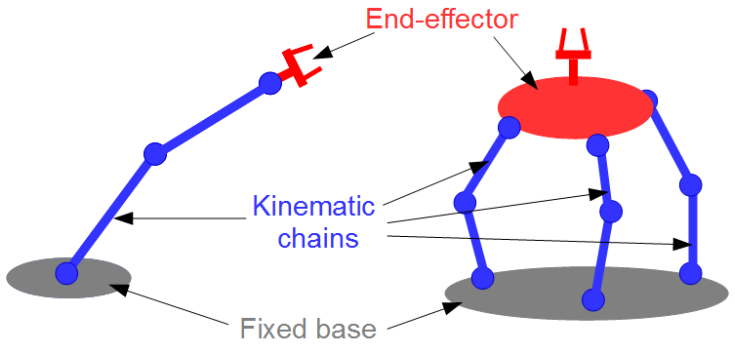


Kinematics

- **Robot dynamics** places **additional constraints** on workspace and **trajectory** due **to mass and force** considerations.
- The mobile robot is also **limited by dynamics**; for instance, a high center of gravity **limits** the practical turning radius of a fast, car-like robot because of the danger of rolling.

- The chief **difference between** a mobile robot and a manipulator arm also introduces a significant challenge **for position estimation** unlike the manipulator that has one **end fixed** to the environment.

- Measuring the position of an arm's end effector is **simply a matter of understanding the kinematics** of the robot and measuring the position of all intermediate joints.



Kinematics

- The manipulator's position is thus always **computable by looking at current** sensor data.
- But a mobile robot is a **self-contained automaton** that can **wholly move** with respect to its environment.

- There is **no direct way** to measure a mobile robot's position **instantaneously**.
- Instead, one must integrate the motion of the robot **over time**.
- Add to this the **inaccuracies** of **motion estimation** due to **slippage** and it is clear that measuring a mobile robot's **position precisely is an extremely challenging task.**

But …

- The process of understanding the motions of a robot begins with the process of describing **the contribution each wheel provides for motion**.
- Each wheel has a **role** in enabling the whole robot to move.
- By the same token, each wheel also **imposes constraints** on the robot's motion; for example, refusing to skid laterally.

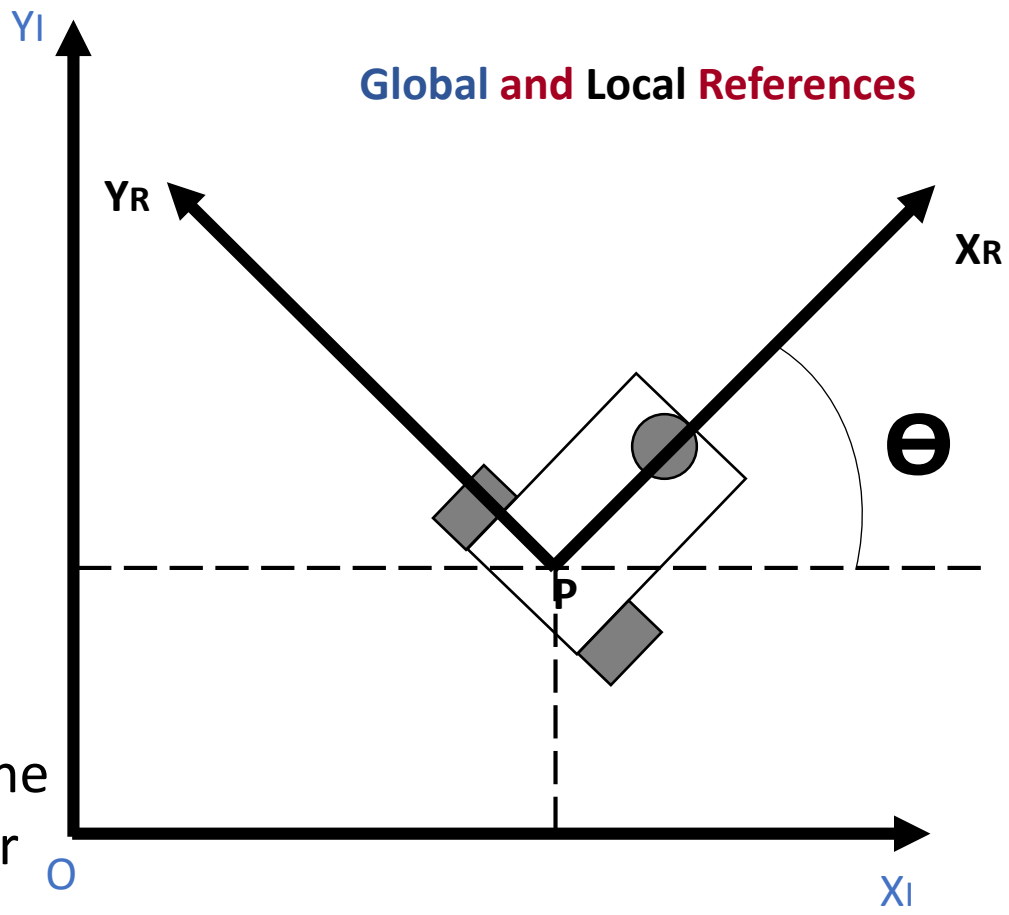
Global and Local frames

Kinematic Models

- Deriving a model for the whole robot's motion is **a bottom-up process**.
- Again, each individual wheel **contributes** to the robot's motion and, at the same time, **imposes constraints** on robot motion.
- Wheels are **tied together based on robot chassis geometry**, and therefore their **constraints combine** to form constraints on the **overall** motion of the robot chassis.
- But the **forces and constraints** of **each** wheel must be **expressed** with respect to a clear and consistent **reference frame**.
- This is particularly **important** in mobile robotics because of its **self-contained** and has mobile nature; a clear mapping between **global and local frames** of reference is required.

Motion Representation

- Assuming the robot as **a rigid body** on **wheels**, operating on a **horizontal plane**.
- The total dimensionality of this robot chassis on the plane is **<u>three</u>**, **two for position in the plane** and o**ne for orientation along the vertical axis**, which is orthogonal to the plane.
- By **robot chassis** it refer only to the rigid body of the robot, **ignoring** the **joints** and degrees of freedom internal to the robot and its wheels.
- Establishing a relationship between the **global reference** frame of the plane and the **local reference** frame of the robot in order to specify the robot's position on **plane**.



We will continue **“Kinematics”**

In Lecture 4